one companion population

companion mass [$M_{\rm Jup}$]

orbital period [yr]

$\mathrm{SNR_{tot}} = 30$

$\mathrm{SNR_{tot}} = 10$

$\mathrm{SNR_{tot}} = 5$

A B C D E F G H I

period localized

period not localized

not significant

no significant peak or acceleration

significant peak, period localized

significant peak or acceleration, period not localized

Gaia DR4 baseline

constant $\mathrm{SNR_{tot}}$ (median over stars)

injected period

Gaia DR4 baseline

null-calibrated detection threshold

$M_c = 15\,M_{\rm Jup},\ M_\star = 0.88\,M_\odot$
$P = 2.48$ yr, $d = 168$ pc
$e = 0.17$, $i = 50.7°$

$M_c = 6.6\,M_{\rm Jup},\ M_\star = 0.68\,M_\odot$
$P = 1.12$ yr, $d = 70.3$ pc
$e = 0.3$, $i = 126°$

$M_c = 40\,M_{\rm Jup},\ M_\star = 1.5\,M_\odot$
$P = 2.62$ yr, $d = 472$ pc
$e = 0.51$, $i = 159°$

$M_c = 12\,M_{\rm Jup},\ M_\star = 0.81\,M_\odot$
$P = 7.07$ yr, $d = 268$ pc
$e = 0.091$, $i = 116°$

$M_c = 33\,M_{\rm Jup},\ M_\star = 0.25\,M_\odot$
$P = 25.2$ yr, $d = 27.3$ pc
$e = 0.76$, $i = 85.8°$

$M_c = 64\,M_{\rm Jup},\ M_\star = 0.88\,M_\odot$
$P = 318$ yr, $d = 441$ pc
$e = 0.56$, $i = 23.9°$

significant via acceleration

$M_c = 0.0096\,M_{\rm Jup},\ M_\star = 0.64\,M_\odot$
$P = 0.102$ yr, $d = 205$ pc
$e = 0.7$, $i = 78.1°$

$M_c = 0.48\,M_{\rm Jup},\ M_\star = 1.2\,M_\odot$
$P = 2.81$ yr, $d = 471$ pc
$e = 0.64$, $i = 133°$

$M_c = 20\,M_{\rm Jup},\ M_\star = 0.99\,M_\odot$
$P = 450$ yr, $d = 377$ pc
$e = 0.53$, $i = 49.6°$

$\Delta\chi^2$

period [yr]